

Intervention Protocol — VPA A-Level Computer Science

A-Level Computer Science Intervention Protocol

A-Level计算机科学干预方案 / Victoria Park Academy

1. Purpose & Scope / 目的与范围

This document sets out the systematic intervention framework for A-Level Computer Science (9618) students at Victoria Park Academy who are identified as struggling or at risk of underachievement. It applies to all Year 12 and Year 13 students following the Cambridge International A-Level Computer Science syllabus.

本文件为维多利亚公园学院A-Level计算机科学（9618）中被识别为学习困难或有成绩不达标风险的学生制定了系统的干预框架。它适用于所有遵循剑桥国际A-Level计算机科学大纲的12年级和13年级学生。

Definition of Intervention / 干预定义: A structured, time-limited, and measurable support strategy designed to close the gap between a student's current performance and the expected standard for A-Level Computer Science. Interventions are proactive (triggered by early warning signs) and reactive (triggered by formal assessment results).

干预是一种结构化、有时间限制且可衡量的支持策略，旨在缩小学生当前表现与A-Level计算机科学预期标准之间的差距。干预是主动的（由早期预警信号触发）和被动的（由正式评估结果触发）。

2. Identification Criteria / 识别标准

Students are identified for intervention through multiple data points. No single indicator is sufficient. 学生通过多个数据点被识别为需要干预。单一指标不足以启动干预。

2.1 Academic Indicators / 学业指标

Indicator / 指标	Threshold / 阈值	Data Source / 数据来源
End-of-topic test score	< 50% (E grade equivalent)	Teacher-assessed tests
Mock examination result	< Grade D (AS) or < Grade C (A2)	Formal mock exams
Programming assignment score	< 60% (fails to meet specification)	Practical coursework
Past paper practice (timed)	< 40% on two consecutive papers	Teacher-marked practice
IGCSE to A-Level transition gap	IGCSE grade B or below entering A-Level	Admissions data
Prerequisite knowledge deficit	Failed summer transition work checklist	Baseline assessment

2.2 Engagement & Behavioural Indicators / 参与度与行为指标

Indicator / 指标	Threshold / 阈值	Data Source / 数据来源
Homework submission rate	< 70% over a half-term	Homework tracker
Class participation	Teacher observation: "rarely contributes"	Teacher log / seating plan

Indicator / 指标	Threshold / 阈值	Data Source / 数据来源
Attendance	< 90% in CS lessons	School MIS system
Lateness to lessons	3+ late arrivals in a half-term	Attendance register
Failure to bring equipment	Consistent lack of laptop/notes	Teacher observation
Self-reported difficulty	Student reports "I don't understand" repeatedly	Student voice / pastoral

2.3 EAL-Specific Indicators / EAL特定指标

Difficulty understanding English-medium technical vocabulary (e.g., "encapsulation," "normalisation," "polymorphism"). 难以理解英语技术词汇（如“封装”、“规范化”、“多态”）。

Can solve problems in Chinese but struggles to express solutions in English pseudocode or prose. 能用中文解决问题，但难以用英语伪代码或散文表达解决方案。

Extended time needed to read and comprehend examination questions. 需要更多时间来阅读和理解考试题目。

Strong mathematical/logical ability masked by language barriers. 强大的数学/逻辑能力被语言障碍掩盖。

Trigger Point / 触发点: A student is flagged for intervention when they meet any two academic indicators OR any one academic indicator plus any two engagement indicators. EAL students may be flagged on EAL-specific criteria alone.

当学生满足任意两项学业指标，或任意一项学业指标加任意两项参与度指标时，将被标记为需要干预。EAL学生可仅根据EAL特定标准被标记。

3. Tier 1: Classroom Support (Universal) / 第一层级：课堂支持（普遍性）

TIER 1 — All Students / 全体学生

High-quality teaching that meets the needs of all learners. Tier 1 is preventative and reduces the need for higher-tier intervention. 满足所有学习者需求的高质量教学。第一层级是预防性的，可减少对更高层级干预的需求。

3.1 Quality-First Teaching Strategies / 优质教学策略

Explicit instruction of technical vocabulary — Every new term is defined, exemplified, and revisited. Maintain a running glossary. 明确教授技术词汇——每个新术语都被定义、举例并反复回顾。维护一个持续更新的词汇表。

Worked examples with faded guidance — Start with fully worked solutions, then gradually remove scaffolding. 带逐步减少指导的示范例题——从完整解答开始，然后逐步减少支架。

Dual coding — Combine diagrams, flowcharts, and code with verbal explanations. 双重编码——将图表、流程图和代码与口头解释结合。

Retrieval practice — Begin every lesson with 5 questions from previous topics. 检索练习——每节课以5个之前主题的问题开始。

Think-Pair-Share — Reduce individual pressure; increase participation. 思考-配对-分享——减少个人压力，增加参与度。

Live coding — Model problem-solving in real-time, verbalising thought processes. 实时编程——实时建模问题解决，口头表达思维过程。

Structured note-taking — Provide Cornell-style note templates for theory lessons. 结构化笔记——为理论课提供康奈尔式笔记模板。

3.2 Differentiation in the Classroom / 课堂差异化

Strategy / 策略	Application in CS / 在CS中的应用
By outcome	All students attempt the same problem; extension challenges for high attainers, structured hints for struggling students.
By task	Struggling students trace a given algorithm; confident students write their own.
By support	Provide sentence starters for written answers; vocabulary banks for EAL students.
By resource	Offer video explanations, interactive simulations, and textbook references for independent review.

3.3 Tier 1 Monitoring / 第一层级监控

Weekly mini-quizzes (5 marks) auto-marked where possible. 每周小测验（5分），尽可能自动评分。

Homework marked within 48 hours with specific, actionable feedback. 作业在48小时内批改，提供具体、可操作的反馈。

Seating plan reviewed half-termly to optimise peer support. 每半学期审视座位安排以优化同伴支持。

Student self-assessment checklists after each topic. 每个主题后的学生自我评估清单。

4. Tier 2: Small Group Intervention (Targeted) / 第二层级：小组干预（针对性）

TIER 2 — 3-6 Students / 3-6名学生

Targeted support for students who have been identified through the criteria in Section 2 but do not require intensive 1-to-1 support. Tier 2 runs for a minimum of 6 weeks. 为通过第2节标准被识别但不需要密集一对一支持的学生提供针对性支持。第二层级至少运行6周。

4.1 Structure / 结构

Frequency: 1 session per week, 45-60 minutes, during lunch or after school. 频率：每周1次，45-60分钟，在午餐时间或放学后。

Group size: 3-6 students with similar needs (e.g., all struggling with programming, or all with theory gaps). 小组规模：3-6名有相似需求的学生（如都在编程上有困难，或都有理论缺口）。

Duration: Minimum 6 weeks; review at Week 3 and Week 6. 持续时间：至少6周；在第3周和第6周进行审查。

Led by: Subject teacher or HLTA with CS specialism. 负责人：学科教师或具有CS专业资格的高级教学助理。

4.2 Tier 2 Intervention Strategies by Topic / 按主题的第二层级干预策略

4.2.1 Programming & OOP / 编程与面向对象编程

Guided programming exercises with partially completed code (scaffolding). 带部分完成代码的引导式编程练习（支架式）。

Peer programming pairs within the group — one student explains, one codes. 小组内的同伴编程配对——一名学生解释，一名学生编码。

Visual OOP modelling using physical objects (e.g., "Car" class with toy cars). 使用实物进行可视化OOP建模（如用玩具车表示"Car"类）。

Code reading exercises — trace and explain what given code does before writing. 代码阅读练习——在编写之前跟踪并解释给定代码的功能。

Debugging clinics — analyse broken code collaboratively. 调试诊所——协作分析有错误的代码。

4.2.2 Theory: Hardware & Architecture / 理论: 硬件与架构

Physical model of the CPU using classroom props (registers = boxes, bus = string). 使用课堂道具制作CPU物理模型(寄存器=盒子, 总线=绳子)。

Step-by-step fetch-decode-execute trace using a large printed diagram. 使用大型打印图表逐步跟踪取指-译码-执行过程。

Assembly language "human computer" simulation — students act as CPU components. 汇编语言"人体计算机"模拟——学生扮演CPU组件。

Comparison cards: match IGCSE concepts to A-Level extensions. 对比卡片: 将IGCSE概念与A-Level扩展内容配对。

4.2.3 Theory: Data Representation / 理论: 数据表示

Binary arithmetic card games — practise addition and subtraction with cards. 二进制算术卡片游戏——用卡片练习加减法。

Floating-point "mantissa and exponent" physical modelling with decimal numbers first. 浮点数"尾数和指数"物理建模, 先用十进制数字。

Colour-by-number bitmap exercises to understand image representation. 按数字填色的位图练习, 以理解图像表示。

Sound wave sampling exercises using Audacity and printed waveforms. 使用Audacity和打印波形进行声波采样练习。

4.2.4 Theory: Algorithms & Complexity / 理论: 算法与复杂度

Sorting algorithm races — students physically sort themselves using different algorithms. 排序算法竞赛——学生用不同算法 physically 排序自己。

Recursion tracing using stack frame diagrams (paper-based before code). 使用栈帧图跟踪递归(先纸质再代码)。

Big-O concept introduced through "how many steps?" counting exercises. 通过"多少步?"计数练习引入大O概念。

Algorithm card matching: match problem to algorithm to complexity. 算法卡片配对: 将问题与算法和复杂度配对。

4.2.5 Theory: Databases / 理论: 数据库

Normalisation workshops using physical cards for entities and attributes. 使用实体和属性的物理卡片进行规范化工作坊。

SQL "sentence building" — construct queries using colour-coded word cards. SQL"句子构建"——使用彩色编码单词卡片构建查询。

E-R diagram construction using whiteboard and magnetic shapes. 使用白板和磁性形状构建E-R图。

4.3 Tier 2 Tracking / 第二层级跟踪

Each Tier 2 student has an intervention record card updated weekly:

TIER 2 INTERVENTION RECORD / 第二层级干预记录	
Student Name:	_____
Group:	_____
Start Date:	_____
Review Date:	_____
Focus Area:	_____

Student Name: _____ Group: _____

Start Date: _____ Review Date: _____

Focus Area: _____

Week 1: Topic covered: _____ Engagement: ☐ High ☐ Med ☐ Low

Week 2: Topic covered: _____ Engagement: ☐ High ☐ Med ☐ Low

Week 3: MID-POINT REVIEW — Progress: ☐ On track ☐ Needs adjustment

Week 4: Topic covered: _____ Engagement: ☐ High ☐ Med ☐ Low

Week 5: Topic covered: _____ Engagement: ☐ High ☐ Med ☐ Low

Week 6: FINAL REVIEW — Exit criteria met: ☐ Yes ☐ No → Move to Tier 3?

Exit Criteria (tick all that apply):

- ☐ Achieved 60%+ on end-of-intervention assessment
- ☐ Homework submission rate improved to $\geq 80\%$
- ☐ Student reports increased confidence
- ☐ Teacher observation: participates actively in class
- ☐ Parent informed of progress

5. Tier 3: 1-to-1 Intensive Support / 第三层级：一对一密集支持

TIER 3 — Individual / 个人

For students who have not made sufficient progress after Tier 2, or who present with significant barriers to learning. Tier 3 is highly individualised and may involve external support. 适用于在第二层级后未取得足够进步，或面临重大学习障碍的学生。第三层级高度个性化，可能涉及外部支持。

5.1 Trigger for Tier 3 / 第三层级触发条件

No measurable progress after 6 weeks of Tier 2 intervention. 经过6周第二层级干预后没有可衡量的进步。

Baseline assessment score $\leq 30\%$ (indicating fundamental gaps). 基线评估分数低于30%（表明存在根本性缺口）。

Student has missed $\geq 20\%$ of lessons due to absence. 学生因缺勤错过了超过20%的课程。

Student has identified SEND needs affecting CS specifically. 学生有已识别的SEND需求，特别影响CS学习。

Student is at risk of failing the A-Level (predicted grade U/E). 学生有A-Level不及格的风险（预测成绩U/E）。

5.2 Structure / 结构

Frequency: 1–2 sessions per week, 30–45 minutes each. 频率：每周1–2次，每次30–45分钟。

Setting: Quiet room, one-to-one with subject teacher or SEN specialist. 环境：安静房间，与学科教师或SEN专家一对一。

Duration: Minimum 8 weeks; continuous review. 持续时间：至少8周；持续审查。

Documentation: Individual Learning Plan (ILP) updated fortnightly. 文档：每两周更新一次的个人学习计划（ILP）。

5.3 Tier 3 Strategies / 第三层级策略

Diagnostic assessment — Identify the exact point of breakdown (e.g., "student understands binary but not two's complement subtraction"). 诊断评估——识别具体崩溃点（如“学生理解二进制但不理解补码减法”）。

Micro-teaching — Break topics into 10-minute chunks with immediate practice. 微教学——将主题分解为10分钟小块，并立即练习。

Overlearning — Repeat core skills until automatic (e.g., binary conversion daily for 2 weeks). 过度学习——重复核心技能直到自动化（如连续2周每天练习二进制转换）。

Multisensory approaches — Use physical manipulatives, colour coding, and movement for kinaesthetic learners. 多感官方法——使用实物操作、颜色编码和动作适合动觉学习者。

Metacognitive scaffolding — "Think aloud" protocols where the student verbalises their reasoning. 元认知支架——“大声思考”协议，学生口头表达他们的推理。

Exam technique coaching — Deconstruct mark schemes; practise writing to the point. 考试技巧辅导——分解评分方案；练习切题写作。

Emotional coaching — Address maths anxiety, coding anxiety, or fixed mindset. 情感辅导——解决数学焦虑、编程焦虑或固定型思维。

5.4 Tier 3 Individual Learning Plan (ILP) Template / 第三层级个人学习计划模板

| TIER 3 INDIVIDUAL LEARNING PLAN (ILP) / 第三层级个人学习计划 |

Student: _____ Year Group: _____ Date: _____

Teacher: _____ SENCo involved: ☐ Yes ☐ No

DIAGNOSTIC FINDINGS / 诊断发现:

TARGETS (SMART) / 目标 (SMART原则):

1. _____ By: _____

2. _____ By: _____

3. _____ By: _____

STRATEGIES / 策略:

☐ Micro-teaching ☐ Overlearning ☐ Multisensory

☐ Exam technique ☐ Emotional coaching ☐ Peer mentoring

☐ Other: _____

RESOURCES / 资源:

☐ Zakonweb theory notes ☐ Cambridge past papers

☐ Python programming tasks ☐ Physical manipulatives

☐ EAL vocabulary bank ☐ Other: _____

REVIEW DATES / 审查日期:

Week 2: _____ Week 4: _____ Week 6: _____ Week 8: _____

OUTCOME / 结果:

☐ Returned to mainstream (Tier 1) ☐ Continued Tier 3

☐ Referral to external support ☐ Other: _____

6. EAL-Specific Interventions / EAL特定干预

English as an Additional Language students face unique challenges in Computer Science, where technical vocabulary is dense and examination responses require precise English. 英语作为附加语言的学生在计算机科学中面临独特挑战，因为技术词汇密集，且考试回答需要精确的英语。

6.1 Vocabulary Support / 词汇支持

Bilingual glossary — Maintain a Chinese-English glossary of all CS terms, updated per topic. 双语词汇表——维护一份中英计算机学术语词汇表，按主题更新。

Word walls — Display key terms with definitions and Chinese translations in the classroom. 词汇墙——在教室展示关键术语及其定义和中文翻译。

Root word analysis — Teach Latin/Greek roots (e.g., "poly-" = many, "morph-" = form). 词根分析——教授拉丁/希腊词根（如"poly-"=多，"morph-"=形态）。

Sentence frames — Provide scaffolded sentence structures for written answers.
句子框架——为书面回答提供支架式句子结构。

6.2 Assessment Accommodations / 评估调整

Accommodation / 调整	Application / 应用	Notes / 备注
Extra time (25%)	All formal assessments and mock exams	Must be applied for via Examinations Officer
Bilingual dictionary	Internal assessments and practice papers	Not permitted in Cambridge exams; use for learning
Pre-teaching vocabulary	Key terms taught 48 hours before new topic	Reduces cognitive load during main lesson
Visual organisers	Mind maps, concept maps for each topic	Reduces reliance on dense English text
Oral assessment option	Where appropriate, verbal explanation recorded	For internal tracking only; builds confidence

6.3 EAL Intervention Strategies by Skill / 按技能的EAL干预策略

Reading Examination Questions / 阅读考试题目

Teach "question deconstruction" — identify command words, topic, and mark allocation.
教授“题目解构”——识别指令词、主题和分数分配。

Provide practice with past paper questions where the English is complex but the CS is simple.
提供历年试题练习，其中英语复杂但CS内容简单。

Use "dual-language highlighting" — student highlights English words they know in one colour, unknown in another.
使用“双语高亮”——学生用一种颜色标记认识的英语单词，另一种颜色标记不认识的。

Writing Pseudocode / 编写伪代码

Accept pseudocode with Chinese comments initially; transition to English over time.
初期接受带中文注释的伪代码；逐渐过渡到英语。

Provide "pseudocode phrasebook" — common structures in English and Chinese side-by-side.
提供“伪代码常用语手册”——常见结构的英语和中文对照。

Focus on logic correctness first, language accuracy second. 先关注逻辑正确性，再关注语言准确性。

Extended Written Answers / 扩展书面回答

Provide paragraph structures: "Point — Evidence — Explain — Link." 提供段落结构：“观点——证据——解释——联系。”

Allow planning in Chinese before writing in English. 允许先用中文规划，再用英语写作。

Use peer review with bilingual students assisting. 使用双语学生协助的同伴评审。

6.4 EAL Parent Communication / EAL家长沟通

See Section 8 for templates. Key principle: communicate in the parent's preferred language where possible. Use translation services or bilingual staff. 见第8节模板。关键原则：尽可能用家长偏好的语言沟通。使用翻译服务或双语员工。

7. Specific Intervention Strategies by A-Level Topic / 按A-Level主题的特定干预策略

7.1 Information Representation (Topic 1.1) / 信息表示

Common difficulty: Students confuse mantissa/exponent, or cannot normalise floating-point numbers.

常见困难：学生混淆尾数/指数，或无法规范化浮点数。

Concrete-pictorial-abstract approach: Start with decimal scientific notation (concrete), move to binary diagrams (pictorial), then formal notation (abstract).

具体-图示-抽象法：从十进制科学记数法（具体）开始，过渡到二进制图示（图示），再到正式表示（抽象）。

Colour-coded worksheets: Mantissa in blue, exponent in red, sign bit in green.
彩色编码工作表：尾数用蓝色，指数用红色，符号位用绿色。

"Spot the error" activities: Give incorrect normalisations; students identify and correct.
"找错误"活动：给出错误的规范化；学生识别并纠正。

7.2 Hardware & Architecture (Topic 1.2) / 硬件与架构

Common difficulty: Students memorise register names but cannot explain the fetch-decode-execute cycle with register contents.

常见困难：学生记住寄存器名称，但无法用寄存器内容解释取指-译码-执行周期。

Role-play simulation: Students take roles (PC, MAR, MDR, CIR, ACC) and physically pass "instructions" around.
角色扮演模拟：学生扮演角色（PC、MAR、MDR、CIR、ACC）并 physically 传递“指令”。

Step-by-step worksheets: Every cycle stage with blank registers to fill in.
分步工作表：每个周期阶段都有空白寄存器需要填写。

Video creation: Students make a 2-minute video explaining the cycle to a peer.
视频制作：学生制作2分钟视频向同伴解释该周期。

7.3 Programming & OOP (Topic 2.2) / 编程与面向对象编程

Common difficulty: Students can write procedural code but cannot design classes or understand inheritance.

常见困难：学生能编写过程式代码，但无法设计类或理解继承。

Real-world analogy first: "A Dog is an Animal" — physical sorting activity before code.
先使用现实类比：“狗是动物”——编码前先进行物理分类活动。

UML diagramming: Draw class diagrams before writing any code. UML图绘制：在编写任何代码之前先绘制类图。

Code completion exercises: Given a class diagram, fill in the Python code. 代码补全练习：给定类图，填写Python代码。

Incremental OOP projects: Start with a single class, add inheritance in Week 2, polymorphism in Week 3.
渐进式OOP项目：从单个类开始，第2周添加继承，第3周添加多态。

7.4 Recursion (Topic 2.1) / 递归

Common difficulty: Students cannot identify base cases or trace recursive calls.

常见困难：学生无法识别基准情形或跟踪递归调用。

"Russian doll" analogy: Each doll is a function call; the smallest doll is the base case.
"俄罗斯套娃"类比：每个套娃是一个函数调用；最小的套娃是基准情形。

Stack frame drawing: Every recursive call gets its own paper "frame" stacked on the desk.
栈帧绘制：每次递归调用都在桌上堆叠自己的纸质“帧”。

Start with mathematical recursion: Factorial and Fibonacci before tree recursion.
从数学递归开始：先学习阶乘和斐波那契，再学习树递归。

7.5 Database Normalisation (Topic 2.3) / 数据库规范化

Common difficulty: Students can identify 1NF but struggle with 2NF, 3NF, and functional dependencies.

常见困难：学生能识别第一范式，但在第二范式、第三范式和函数依赖方面有困难。

"Unnormalised to normalised" card sort: Physical cards representing data; students sort into tables.
"非规范化到规范化"卡片分类：代表数据的物理卡片；学生将其分类到表中。

Functional dependency mapping: Draw arrows between attributes before formalising.
函数依赖映射：在形式化之前先绘制属性之间的箭头。

Repeated practice: 10 different unnormalised schemas; normalise to 3NF each.
重复练习：10个不同的非规范化模式；每个都规范化到第三范式。

7.6 Boolean Algebra & Logic Circuits (Topic 1.2) / 布尔代数与逻辑电路

Common difficulty: Students cannot apply De Morgan's laws or simplify using Karnaugh maps.

常见困难：学生无法应用德摩根定律或使用卡诺图简化。

Truth table verification: Prove De Morgan's by completing both sides of the truth table.
真值表验证：通过完成真值表两边来证明德摩根定律。

K-map colouring: Use highlighters to identify groups on printed K-maps before drawing.
卡诺图着色：在绘制之前使用荧光笔在打印的卡诺图上标识组。

Circuit simulation software: Use Logisim (free) to build and test circuits visually.
电路仿真软件：使用Logisim（免费）可视地构建和测试电路。

8. Parent Communication Templates / 家长沟通模板

8.1 Initial Concern Letter / 初步关注函

Dear [Parent Name],

I am writing to inform you that [Student Name] has been identified as needing additional support in A-Level Computer Science. This is a proactive intervention designed to ensure [Student Name] achieves their full potential.

Current concerns:

- ☐ [Specific assessment score / engagement issue]
- ☐ [Specific topic difficulty]

Planned support:

- ☐ [Tier 1/2/3 intervention details]
- ☐ [Frequency and duration]

We would welcome the opportunity to discuss this with you. Please contact me to arrange a brief meeting or phone call.

Yours sincerely,

[Teacher Name]

Head of Computer Science

Victoria Park Academy

尊敬的[家长姓名]：

我写信通知您，[学生姓名]在A-Level计算机科学中被识别为需要额外支持。这是一项预防性干预，旨在确保[学生姓名]充分发挥

其潜力。

当前关注点：

☐ [具体评估分数/参与度问题]

☐ [具体主题困难]

计划支持：

☐ [第一/二/三层级干预详情]

☐ [频率和持续时间]

我们期待与您讨论此事。请联系我安排简短会面或电话沟通。

此致

[教师姓名]

计算机科学学科主管

维多利亚公园学院

8.2 Progress Update / 进展更新

Dear [Parent Name],

Update on [Student Name]'s intervention progress:

Intervention focus: _____

Duration: ____ weeks completed

Progress indicators:

☐ Assessment scores improved from ____% to ____%

☐ Homework submission rate: ____%

☐ Class participation: ☐ Increased ☐ Stable ☐ Decreased

☐ Student confidence: ☐ Increased ☐ Stable ☐ Decreased

Next steps:

Thank you for your continued support.

[Teacher Name]

尊敬的[家长姓名]：

[学生姓名]的干预进展更新：

干预重点：_____

持续时间：已完成__周

进展指标：

☐ 评估分数从__%提高到__%

☐ 作业提交率：__%

☐ 课堂参与度： ☐ 提高 ☐ 稳定 ☐ 下降

☐ 学生信心： ☐ 提高 ☐ 稳定 ☐ 下降

下一步：

感谢您持续的支持。

[教师姓名]

8.3 EAL-Specific Parent Letter / EAL特定家长函

Dear [Parent Name],

[Student Name] is making good progress in Computer Science, but we have identified that language is a barrier to demonstrating their full understanding in written assessments.

Support being provided:

- ☐ Bilingual vocabulary glossary for all topics
- ☐ Extra time in assessments (25%)
- ☐ Pre-teaching of key terms before each topic
- ☐ Visual organisers and concept maps

How you can help at home:

- ☐ Discuss CS topics in [home language] to reinforce understanding
- ☐ Encourage [Student Name] to explain concepts aloud
- ☐ Review the bilingual glossary together

We are confident that with this support, [Student Name] will overcome the language barrier and achieve highly.

[Teacher Name]

尊敬的[家长姓名]:

[学生姓名]在计算机科学方面进步良好,但我们发现语言是其在书面评估中充分展示理解的障碍。

提供的支持:

- ☐ 所有主题的双语词汇表
- ☐ 评估中的额外时间(25%)
- ☐ 每个主题前的关键术语预教
- ☐ 可视化组织工具和概念图

您在家可以如何帮助:

- ☐ 用[家庭语言]讨论CS主题以加强理解
- ☐ 鼓励[学生姓名]大声解释概念
- ☐ 一起复习双语词汇表

我们相信,在这些支持下,[学生姓名]将克服语言障碍并取得优异成绩。

[教师姓名]

9. Tracking & Review / 跟踪与审查

9.1 Intervention Tracker (Whole Cohort) / 干预跟踪表(整个年级)

Student / 学生	Year / 年级	Tier / 层级	Focus / 重点	Start Date / 开始日期	Review Date / 审查日期	Progress / 进展	Exit Date / 退出日期
Example Student A	12	Tier 2	OOP programming	15/09/2025	15/11/2025	On track	—
Example Student B	13	Tier 3	Data representation	01/10/2025	01/12/2025	Needs adjustment	—

9.2 Review Schedule / 审查时间表

Weekly: Teacher reviews intervention group progress; updates record cards. 每周：教师审查干预小组进展；更新记录卡。

Half-termly: Department meeting reviews all intervention data; adjusts groupings. 每半学期：部门会议审查所有干预数据；调整分组。

Termly: SLT review of intervention impact; resource allocation decisions. 每学期：SLT审查干预影响；资源分配决策。

Annually: Full protocol review; update based on examination results and student voice. 每年：全面协议审查；根据考试成绩和学生反馈更新。

9.3 Success Metrics / 成功指标

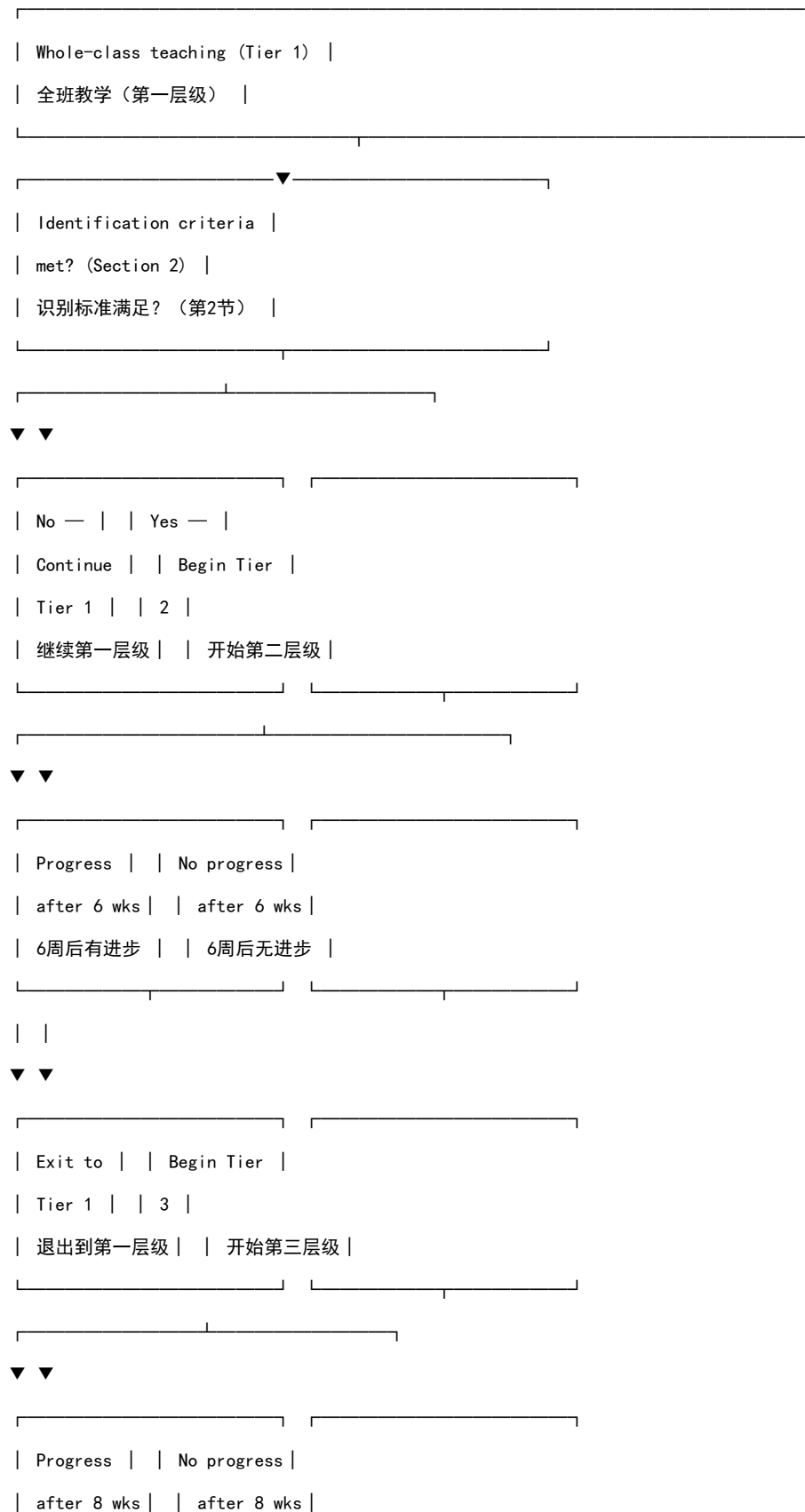
Metric / 指标	Target / 目标	Measurement / 测量
Students exiting intervention	≥70% exit within 8 weeks	Intervention tracker
Grade improvement	≥1 grade improvement for Tier 3 students	Mock vs. final grade
Homework submission	≥80% for all intervention students	Homework tracker
Student confidence	Self-reported increase on exit survey	Exit questionnaire
Parent satisfaction	≥80% positive feedback	Parent survey

10. Roles & Responsibilities / 角色与职责

Role / 角色	Responsibilities / 职责
Head of Computer Science	Overall protocol ownership; quality assurance; Tier 3 delivery; staff training; parent communication for Tier 3.
Classroom Teacher	Tier 1 quality-first teaching; identification of students; Tier 2 delivery; weekly tracking updates.
HLTA / Teaching Assistant	Support Tier 2 sessions; resource preparation; 1-to-1 support under teacher direction.
SENCo	Advise on SEND-specific strategies; liaise with external agencies; ensure access arrangements.
EAL Coordinator	Bilingual resource development; EAL-specific strategy advice; parent communication in home language.
Head of Year / Pastoral	Attendance monitoring; emotional support; communication with parents; barrier identification.
Student	Attend all intervention sessions; complete assigned work; communicate difficulties; engage with feedback.
Parent	Support at home; attend meetings; reinforce vocabulary; encourage independent study.

11. Appendices / 附录

Appendix A: Intervention Flowchart / 附录A: 干预流程图



8周后有进步	8周后无进步
▼ ▼	
Exit to	External
Tier 1/2	referral
退出到第一/二层级	外部转介

Appendix B: Student Self-Assessment Checklist / 附录B: 学生自我评估清单

Students complete this after each topic. 学生在每个主题后完成此清单。

Topic: _____ Date: _____

Rate your confidence (1 = not confident, 5 = very confident):

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 I can explain this topic to a peer.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 I can answer past paper questions on this topic.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 I can write code related to this topic.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 I understand the key vocabulary.

What I found difficult: _____

What I need help with: _____

Victoria Park Academy | A-Level Computer Science (9618) | Intervention Protocol

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